Chapter 16
Acid-Base Equilibria

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Definition of Acids & Bases

16.1 Arrhenius Definition:

$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$

Acid ⇨ **produces H^+ in aqueous solutions**

$NaOH(aq) \rightarrow Na^+(aq) + OH^-(aq)$

Base ⇨ **produces OH^- in aqueous solutions**

$CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$

Acetic acid — Acetate ion

$NH_3(g) + H_2O \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

ammonia — Ammonium ion

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Definition of Acids & Bases

16.2 Brønsted-Lowry Definition:

remove H^+

$$HF(g) + NH_3(g) \rightleftharpoons NH_4^+ + F^-$$

add H^+

$HF(g)$	$NH_3(g)$	NH_4^+	F^-
Acid	Base	Conjugate Acid	Conjugate Base

Acid: proton (H^+) donor

Base: proton (H^+) acceptor

NH_3/NH_4^+ → conjugate acid-base pair

HF/F^- → conjugate acid-base pair

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Definition of Acids & Bases

16.2 Brønsted-Lowry Definition:

remove H^+

$$\underbrace{H_2O(aq)}_{\text{Acid}} + \underbrace{NH_3(aq)}_{\text{Base}} \rightleftharpoons \underbrace{NH_4^+(aq)}_{\text{Conjugate Acid}} + \underbrace{OH^-(aq)}_{\text{Conjugate Base}}$$

add H^+

Acid: proton (H^+) donor

Base: proton (H^+) acceptor

Conjugate Acid

Conjugate Base

NH_3/NH_4^+ → conjugate acid-base pair

H_2O/OH^- → conjugate acid-base pair

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$$\overbrace{HNO_2(aq) + \underbrace{H_2O(L) \rightleftharpoons H_3O^+}_{\text{add } H^+}(aq) + NO_2^-(aq)}^{\text{remove } H^+}$$

Acid — proton (H^+) donor → must have H^+ to donate

Base — proton (H^+) acceptor → must have nonbonding e-pair to accept that proton.

Conjugate Acid

Conjugate Base

H_2O → Amphoteric, can react as an acid or as a base.

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Sample Exercise

What is the conjugate Base of:

acid $\xrightarrow{-H^+}$	conjugate base
HCO_3^-	CO_3^{2-}
OH^-	O^{2-}
NH_3	NH_2^-
$H_2PO_4^-$	HPO_4^{2-}
PO_4^{3-}	??

Sample Exercise

What is the conjugate Acid of:

base $\xrightarrow{+ H^+}$	Conjugate acid
HCO_3^-	H_2CO_3
OH^-	H_2O
F^-	HF
PO_4^{3-}	HPO_4^{2-}
CH_4	??

Relative Strengths of Acids & Bases

Acid strength: its ability to loose H^+

Base strength: its ability to gain H^+

The stronger an acid, the weaker its conjugate base, and the stronger a base , the weaker its conjugate acid.

Determined by %ionization or *K*.
If at the same *T* and concentration:

$$HA + H_2O \rightleftharpoons H_3O^+ + A^- \qquad 20\%$$

$$HX + H_2O \rightleftharpoons H_3O^+ + X^- \qquad 10\%$$

HA stronger acid than HX (larger K_a)

Acids and Bases categories

1. A strong acid (conjugate base has NEGLIGIBLE BASICITY).
   - completely transfer its proton to H_2O
   - (100% dissociation).

$$HCl + H_2O \rightarrow H_3O^+ + Cl^- \qquad 100\%$$

$\underbrace{HCl, HBr, HI, HNO_3, HClO_4, HClO_3,}_{\text{monoprotic acids}}$ H_2SO_4

monoprotic acids → one acidic proton

H_2SO_4 → diprotic acids → more than one acidic protons

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2. A weak acid (the conjugate base is a WEAK BASE).

- partially dissociates in H_2O (< 100%).

$$CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$$

ACID	$\xrightarrow{-H^+}$	Base
HCl		Cl^-
H_2SO_4		HSO_4^-
H_3O^+		H_2O
HSO_4^-		SO_4^{2-}
HF		F^-
CH_3COOH		CH_3COO^-
NH_4^+		NH_3
H_2O		OH^-

3. A substance with negligible acidity (the conjugate base is a STRONG BASE).

- STRONG Base reacting completely with H_2O (100% dissociates).

<u>STRONG Bases:</u>

1. Ionic hydroxides of alkali metals such as NaOH, KOH.

2. Ionic hydroxides of heavier alkaline earth metals such as $Sr(OH)_2$, $Ca(OH)_2$

3. Ionic oxides of alkali and heavier alkaline earth metals such as Na_2O, CaO.

$$Na_2O \rightarrow Na^+ + O^{2-}$$

$$O^{2-} + H_2O \rightarrow OH^- + OH^-$$

STRONG Bases:

4. Hydrides: NaH, CaH_2

$NaH\ (s) \rightarrow Na^{+}\ (aq) + H^{-}\ (aq)$

$H^{-}\ (aq) + H_2O \rightarrow H_2\ (g) + OH^{-}\ (aq)$

5. Nitrides: Na_3N, Ca_3N_2

$Na_3N\ (s) \rightarrow 3Na^{+}\ (aq) + N^{3-}\ (aq)$

$N^{3-}(aq) + 3H_2O \rightarrow NH_3\ (aq) + 3OH^{-}\ (aq)$

	Acid ($- H^+$ →)	Base	
strong acids	HCl	Cl^-	negligible basicity
	H_2SO_4	HSO_4^-	
	H_3O^+	H_2O	
weak acids	HSO_4^-	SO_4^{2-}	weak bases
	HF	F^-	
	CH_3COOH	CH_3COO^-	
	NH_4^+	NH_3	
	H_2O	OH^-	
negligible acidity	OH^-	O^{2-}	strong bases
	H_2	H^-	

Increasing acid strength ↑

Increasing base strength ↓

Predict the direction of the following reaction.

$$CH_3COOH(aq) + NH_3(aq) \rightleftharpoons CH_3COO^-(aq) + NH_4^+(aq)$$

stronger acid → weaker acid

proceeds to right

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16.3 Autoionization of Water

$$H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+ + OH^-$$

Hydronium ion (H_3O^+) Hydroxyl ion (OH^-)

$$K_c = \frac{[H_3O^+][OH^-]}{[H_2O]^2}$$

$[H_2O]$ = constant

$$K_c\,[H_2O] = \boxed{K_w = [H_3O^+][OH^-]}\ \boxed{= 1.0 \times 10^{-14}}$$

Ion-product constant (K_w) only at 25°C (1.0×10^{-14})

$$H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+ + OH^-$$

Solution

$[H^+] = [OH^-]$

$[H^+] = [OH^-] = 1.0 \times 10^{-7}$ 25°C

neutral

$[H^+] > [OH^-]$

$[H^+] > 1.0 \times 10^{-7}$ 25°C

acidic

$[H^+] < [OH^-]$

$[H^+] < 1.0 \times 10^{-7}$ 25°C

basic

What is the concentration of OH^- ions in a HCl solution whose hydrogen ion concentration is 1.3 *M* at 25°C?

$$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$$

	HCl(aq)	H^+(aq)	Cl^-(aq)
Initially, i	1.3 M	0.00 M	0.00 M
change, Δ_c	-0.13 M	+1.3 M	+1. 3 M
Finally, f	0.00 M	1.3 M	1.3 M

$$K_w = [H^+][OH^-] = 1.0 \times 10^{-14}$$

$$[OH^-] = \frac{K_w}{[H^+]} = \frac{\boxed{1.0 \times 10^{-14}}}{1.3}$$

$$[OH^-] = 7.7 \times 10^{-15}\ M$$

Is the solution acidic or basic?

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16.4 pH Scale

power of hydrogen

$\boxed{pH = -\log[H^+]}$ ⟹ $[H^+] = 10^{-pH}$

$\boxed{pOH = -\log[OH^-]}$ ⟹ $[OH^-] = 10^{-pOH}$

$[H^+][OH^-] = K_w = 1.0 \times 10^{-14}$ at 25°C

$\log[H^+] + \log[OH^-] = \log K_w = \log 1.0 \times 10^{-14}$

$-\log[H^+] + (-\log[OH^-]) = -\log K_w = -\log 1.0 \times 10^{-14}$

$\boxed{pH + pOH = pK_w} = 14$

Only at 25°C

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$$H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+ + OH^-$$

Solution			Solution
??	⟸	$[H^+] = [OH^-]$	neutral
7	⟸	$[H^+] = [OH^-] = 1.0 \times 10^{-7}$ 25°C	
??	⟸	$[H^+] > [OH^-]$	acidic
< 7	⟸	$[H^+] > 1.0 \times 10^{-7}$ 25°C	
??	⟸	$[H^+] < [OH^-]$	basic
> 7	⟸	$[H^+] < 1.0 \times 10^{-7}$ 25°C	

As [H+] increases pH decreases

Can the pH be < 0?

What is the pH of 2 M HCl?

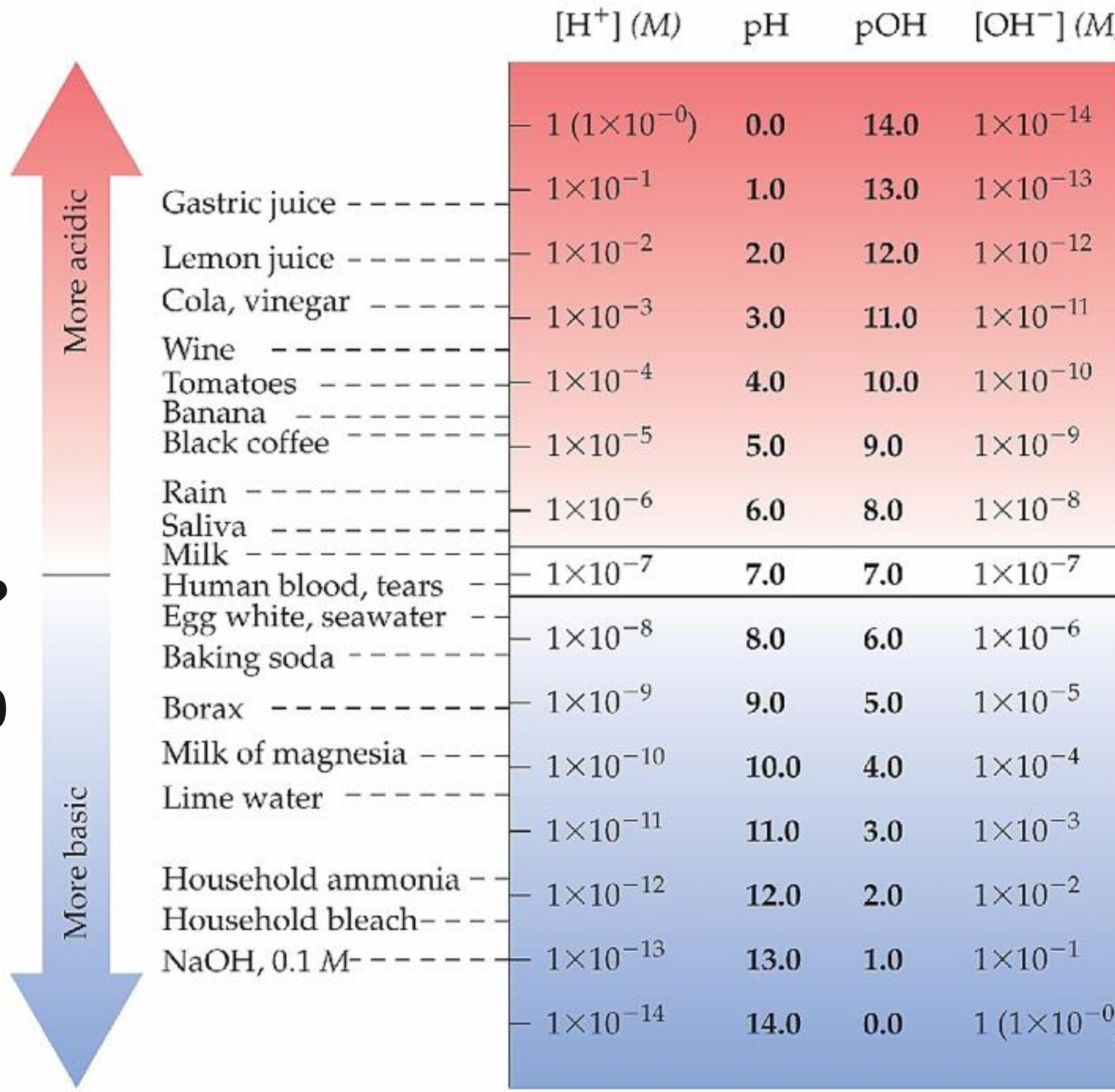
pH = -log 2 = - 0.3

Can the pH be > 14?

What is the pH of 10 M NaOH?

pOH = -log 10 = - 1

pH = 14 - (- 1) = 15



What is the pH of a solution that is 0.10 M in HCl?

$$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$$

	$HCl(aq)$	$H^+(aq)$	$Cl^-(aq)$
Initially, i	0.10 M	0.00 M	0.00 M
change, Δ_c	-0.10 M	+0.10 M	+0.10 M
Finally, f	0.00 M	0.10 M	0.10 M

$$pH = -\log[H^+] = -\log(0.10) = 1.0$$

$$pOH = 14 - pH = 14 - 1 = 13$$

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What is the pH of 2×10^{-3} *M* HNO_3 solution?

HNO_3 is a strong acid → 100% ionization

	$HNO_3(aq)$	$\rightarrow$	$H^+(aq)$	+	$NO_3^-(aq)$
Initially, i	2×10^{-3}		0.0		0.0
change, Δ_c	$- 2 \times 10^{-3}$		$+ 2 \times 10^{-3}$		$+ 2 \times 10^{-3}$
Finally, f	0.0		2×10^{-3}		2×10^{-3}

$$pH = -\log[H^+]$$

$$pH = -\log(2 \times 10^{-3})$$

$$pH = 2.7 \qquad pOH = 11.3$$

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What is the pH of a solution that is 0.10 M in NaOH?

$$NaOH(aq) \rightarrow Na^+(aq) + OH^-(aq)$$

	NaOH(aq)	$Na^+(aq)$	$OH^-(aq)$
initially	0.10 M	0.00 M	0.00 M
finally	0.00 M	0.10 M	0.10 M

$$pOH = -\log[OH^-] = -\log(0.10) = 1.0$$

$$pH = 14 - pOH = 14 - 1 = 13$$

OR

$$[H^+] = \frac{K_w}{[OH^-]} = \frac{1.0 \times 10^{-14}}{0.10} = 1.0 \times 10^{-13}\ M$$

$$pH = -\log 1.0 \times 10^{-13} = 13$$

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What is the pH of a solution that is 0.05 M in $Ca(OH)_2$?

$$Ca(OH)_2(aq) \rightarrow Ca^{2+}(aq) + 2OH^-(aq)$$

	$Ca(OH)_2$	Ca^{2+}	OH^-
initially	0.05 M	0.00 M	0.00 M
finally	0.00 M	0.05 M	0.10 M = 2 x 005

$$pOH = -\log[OH^-] = -\log(0.10) = 1.0$$

$$pH = 14 - pOH = 14 - 1 = 13$$

OR

$$[H^+] = \frac{K_w}{[OH^-]} = \frac{1.0 \times 10^{-14}}{0.10} = 1.0 \times 10^{-13}\ M$$

$$pH = -\log 1.0 \times 10^{-13} = 13$$

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Calculate the pH of 1.8×10^{-2} *M* $Ba(OH)_2$ solution.

$Ba(OH)_2$ is a strong base 100% ionization

	$Ba(OH)_{2(s)} \rightarrow$	$Ba^{2+}(aq)$ +	$2OH^{-}(aq)$
i	1.8×10^{-2}	0.0	0.0
Δ_c	$- 1.8 \times 10^{-2}$	$+ 1.8 \times 10^{-2}$	$+ 2(1.8 \times 10^{-2})$
f	0.0	1.8×10^{-2}	$2(1.8 \times 10^{-2})$

$$pOH = -\log (0.036) = 1.4$$

$$pH = 14 - 1.4 = 12.6$$

Calculate the pH of 0.10 M Ca_3N_2

$Ca_3N_2\,(aq) \rightarrow 3Ca^{2+}\,(aq) + 2N^{3-}\,(aq)$

0.10 M **0.20 M**

$N^{3-}(aq) + 3H_2O \rightarrow NH_3\,(aq) + 3OH^{-}\,(aq)$

0.20 M **0.60 M**

pOH = -log(0.60) = 0.22

pH = 14 – 0.22 = 13.78

$\boxed{\text{The pH of } 1.0 \times 10^{-8}\ M \text{ HCl(aq) solution at } 25^oC \text{ is -------}}$

A. 8 B. 7 C. 6 D. 4

$$HCl(aq) + H_2O \rightarrow H_3O^+(aq) + Cl^-(aq)$$

Initially, i 1.0×10^{-8} 0.00 M

Finally, f 0.00 M $\boxed{1.0 \times 10^{-8}}$

$$pH = -\log(1.0 \times 10^{-8}) \Rightarrow \boxed{pH = \cancel{8.0}}$$

Should consider $[H^+]$ from water

$$H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+ + OH^-$$

$$\boxed{1.0 \times 10^{-7}\ M}$$

$$[H_3O^+]_{total} = 1.0 \times 10^{-7}\ M + 1.0 \times 10^{-7}\ M$$

$$[H_3O^+]_{total} = 1.1 \times 10^{-7}\ M$$

$$pH = -\log(1.1 \times 10^{-7}) \Rightarrow \boxed{pH \cong 7.0}$$

16.6 Weak Acids

$$HA(aq) + H_2O(aq) \rightleftharpoons A^-(aq) + H_3O^+(aq)$$

weak acid

<100% ionization → mixture contains H_3O^+, A^-, and HA

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

acid ionization constant

Indicate the tendency of the acid to ionize in water: the larger the value of the K_a the stronger the acid

Values of K_a for Some Common Monoprotic Acids

Acid	K_a	Conjugate base
HSO_4^-	1.2×10^{-2}	SO_4^{-2}
HF	7.2×10^{-4}	F^-
CH_3COOH	1.8×10^{-5}	CH_3COO^-
HCN	6.2×10^{-10}	CN^-
NH_4^+	1.6×10^{-10}	NH_3

$$HSO_4^- > HF > CH_3COOH > HCN > NH_4^+$$

highest K_a

strongest acid

lowest K_a

weakest acid

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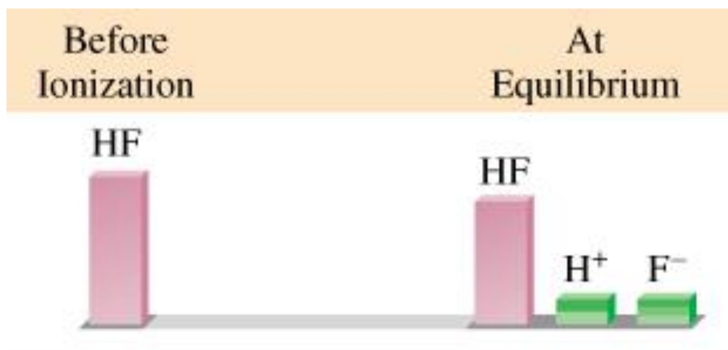
What is the pH of a 0.50 *M* HF solution? (at 25°C).

$K_a = 7.1 \times 10^{-4}$

	$HF_{(aq)}$	$\rightleftharpoons$	$H^+_{(aq)}$	+	$F^-_{(aq)}$
i M	0.50		0.0		0.0
Δ *M*	- x		+ x		+ x
eq M	0.50 - x		x		x



$$K_a = \frac{[H^+][F^-]}{[HF]}$$

$$7.1 \times 10^{-4} = \frac{x^2}{0.50 - x}$$

Assumption: If $x << 0.50\ M$

Then $0.50 - x \approx 0.50\ M$

$$7.1 \times 10^{-4} = \frac{x^2}{0.50 - \cancel{x}} \Rightarrow x^2 = (7.1 \times 10^{-4})(0.50)$$

$$[H^+]_{eq} = x = 0.019\ M \Leftarrow x = \sqrt{(7.1x10^{-4})(0.5)}$$

$$[F^-]_{eq} = x = 0.019\ M$$

$$\boxed{x = \sqrt{(K_a)([ACID]_i)}}$$

only if %ionization < 5%

$$[HF]_{eq} = 0.50 - x = 0.50 - 0.019 = 0.48\ M$$

$$pH = -\log [H^+] = -\log (0.019) \Rightarrow pH = 1.72$$

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Calculate the percentage of the acid that is ionized.

$$HA \rightleftharpoons H^+ + A^-$$

$$\%\text{ionization} = \frac{[H^+]}{[HA]_i} \times 100\%$$

	$HF_{(aq)}$	$\rightleftharpoons$	$H^+_{(aq)}$	+	$F^-_{(aq)}$
i M	0.50		0.0		0.0
eq M	0.50 - x		x		x
	0.48		0.019		0.019

$$\%\text{ionization} = \frac{0.019}{0.50} \times 100\%$$

$$= 3.8\% < 5\% \quad \text{OK}$$

What is the pH of a 0.050 *M* HF solution? (at 25°C).

$K_a = 7.1 \times 10^{-4}$

	$HF_{(aq)}$	$\rightleftharpoons$	$H^+_{(aq)}$	+	$F^-_{(aq)}$
i *M*	0.050		0.0		0.0
Δ *M*	- x		+ x		+ x
eq *M*	0.050 - x		x		x

$$K_a = \frac{[H^+][F^-]}{[HF]} \Rightarrow 7.1 \times 10^{-4} = \frac{x^2}{0.050 - x}$$

Assume x << 0.05 *M*

$$x = 0.006\ M \Leftarrow 7.1 \times 10^{-4} = \frac{x^2}{0.050}$$

$$\frac{0.006}{0.050} \times 100\% = 12\% \quad > 5\% \quad \text{not OK}$$

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Must solve quadratic equation

$$7.1 \times 10^{-4} = \frac{x^2}{\boxed{0.050 - x}}$$

$$x^2 + 7.1 \times 10^{-4}x - 3.6 \times 10^{-5} = 0$$

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$\boxed{\cancel{x = -0.0064\ M}}$

$x = 0.0057\ M$

$[H^+] = x = 0.0057\ M \Rightarrow pH = -\log(0.0057) = 2.24$

$$\%\text{ionization} = \frac{0.0057}{0.050} \times 100\% = 11.4\%$$

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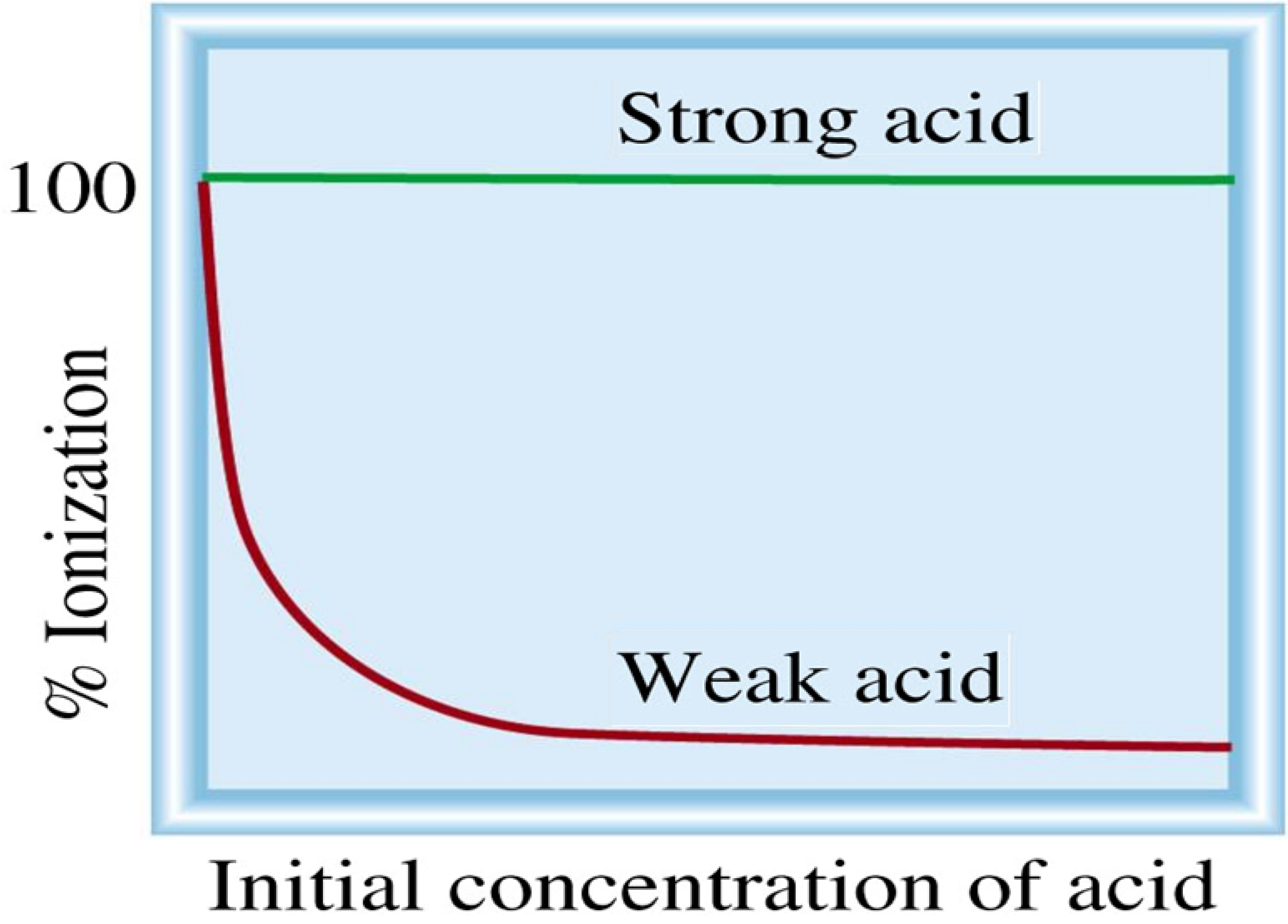
Percent Ionization

$[HF]_i$	0.50 *M*	0.40 *M*	0.050 *M*	
$[H^+]$	0.019	0.017	0.0057	⬇
pH	1.72	1.77	2.24	⬆
% ionization	3.8%	4.3%	11.4%	⬆

As the concentration of a weak acid solution decreases, $[H^+]$ decreases, the pH increases, and the percent ionization increases



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The **pH** of a 0.060 *M* weak monoprotic acid is **3.44**. Calculate the K_a of the acid.

	HA	$\rightleftharpoons$	H^+	+	A^-
i	0.060		0.0		0.0
Δ	-x		+x		+x
eq	0.060 - x		x		x

$$K_a = \frac{[H^+][A^-]}{[HA]} = \frac{x^2}{0.060 - x}$$

$$x = [H^+] = 10^{-pH} = 10^{-3.44} = 3.6 \times 10^{-4}\ M$$

$$K_a = \frac{(3.6 \times 10^{-4})\ (3.6 \times 10^{-4})}{0.060 - 3.6 \times 10^{-4}}$$

$$K_a = 2.2 \times 10^{-6}$$

Polyprotic Acids

Produces more than one H^+ in solution.

$$H_2SO_3 \rightleftharpoons H^+ + HSO_3^- \qquad K_{a1} = 1.7 \times 10^{-2}$$

$$HSO_3^- \rightleftharpoons H^+ + SO_3^{2-} \qquad K_{a2} = 6.4 \times 10^{-8}$$

Easier to remove H^+ from a neutral species than from a negatively charged ion.

ALWAYS: $K_{a1} > K_{a2} > K_{a3} > \ldots$

In solution, consider $[H^+]$ only from first ionization if $K_{a1}/K_{a2} \geq 10^3$.

Calculate the concentration of all species present and the pH in a 0.040 M H_2CO_3 acid solution. $K_{a1} = 4.3 \times 10^{-7}$; $K_{a2} = 5.6 \times 10^{-11}$

	H_2CO_3	$\rightleftharpoons$	H^+	+	HCO_3^-
i	0.040		0		0
Δ	- x		+ x		+ x
eq	0.040 - x		x		x

$$K_{a1} = \frac{[H^+][HCO_3^-]}{[H_2CO_3]} \qquad 4.3 \times 10^{-7} = \frac{x^2}{0.040 - \cancel{x}}$$

Using approximation: $\boxed{x = 1.3 \times 10^{-4}\ M}$

Check: $\dfrac{1.3 \times 10^{-4}}{0.040} \times 100\% = 0.33\%$ OK

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$$x = [H^+] = [HCO_3^-] = 1.3 \times 10^{-4}\ M$$

$$pH = -\log(1.3 \times 10^{-4}) = 3.89$$

HCO_3^-	$\rightleftharpoons$	H^+	+	CO_3^{2-}
1.3×10^{-4}		1.3×10^{-4}		0
$-\ y$		$+\ y$		$+\ y$
$1.3 \times 10^{-4} - y$		$1.3 \times 10^{-4} + y$		y

$$K_{a2} = \frac{[H^+][CO_3^{2-}]}{[HCO_3^-]}$$

$$5.6 \times 10^{-11} = \frac{(\cancel{1.3 \times 10^{-4}} + \cancel{y})(y)}{(\cancel{1.3 \times 10^{-4}} - \cancel{y})}$$

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$$\boxed{y = [CO_3^{2-}] = 5.6 \times 10^{-11}\ M = K_{a2}}$$

$$\boxed{y = [H^+] \text{ from } 2^{nd} \text{ ionization} = 5.6 \times 10^{-11}\ M}$$

16.7 Weak Bases

$$:B(aq) + H_2O(aq) \rightleftharpoons HB^+(aq) + OH^-(aq)$$

weak base

<100% ionization → mixture contains HB^+, OH^-, and ;B

$$K_b = \frac{[HB^+][OH^-]}{[:B]}$$

base ionization constant

Indicate the tendency of the base to ionize in water: the larger the value of the K_b the stronger the base and the weaker the conjugate acid.

Values of K_b for Some Common Bases

Base		K_b	Conjugate acid
NH_3	ammonia and its derivatives	1.8×10^{-5}	NH_4^+
CH_3NH_2		4.4×10^{-4}	$CH_3NH_3^+$
F^-	conjugate bases for weak acids	1.5×10^{-11}	HF
CH_3COO^-		5.6×10^{-10}	CH_3COO^-

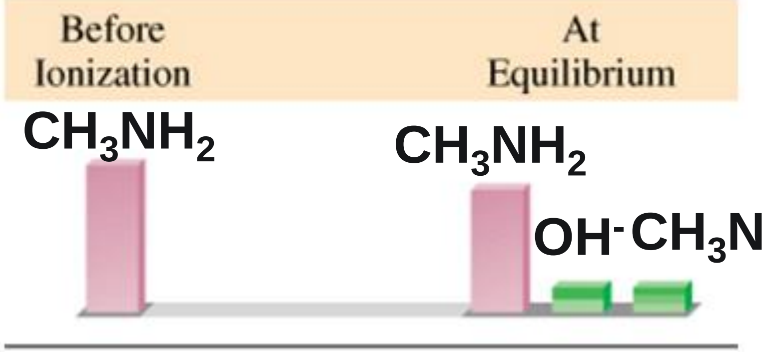
$$CH_3NH_2 > NH_3 > CH_3COO^- > F^-$$

highest K_b
strongest base

lowest K_b
weakest base

Calculate the pH of a 0.20 M solution of methylamine, CH_3NH_2. $K_b = 4.4 \times 10^{-4}$.

	CH_3NH_2 + H_2O	$\rightleftharpoons$	$CH_3NH_3^+$	+ OH^-
i	0.20		0	0
Δ	- x		+ x	+ x
eq	0.20 - x		x	x



$$K_b = \frac{[CH_3NH_3^+][OH^-]}{[CH_3NH_2]} \Rightarrow 4.4 \times 10^{-4} = \frac{x^2}{0.20 - x}$$

Assume x << 0.20 $\Rightarrow$ $4.4 \times 10^{-4} = \dfrac{x^2}{0.20 - \cancel{x}}$

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$$4.4 \times 10^{-4} = \frac{x^2}{0.20 - \cancel{x}} \Rightarrow x^2 = (4.4 \times 10^{-4})(0.20)$$

$$[OH^-]_{eq} = x = 0.0094\,M \Leftarrow x = \sqrt{(4.4 x 10^{-4})(0.2)}$$

$$CH_3NH_3^+ = x = 0.0094\ M$$

$$\boxed{x = \sqrt{(K_b)([BASE]_i)}}$$

only if %ionization < 5%

$$CH_3NH_2 = 0.2 - x \quad = 0.20 - 0.0094 = 0.196\ M$$

$$pOH = -\log(9.4 \times 10^{-3}) \Rightarrow pOH = 2.03$$

$$pH = 14 - 2.00 \Rightarrow pH = 11.97$$

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Calculate the percentage of the base that is ionized.

$$:B(aq) + H_2O(aq) \rightleftharpoons HB^+(aq) + OH^-(aq)$$

$$\%\text{ionization} = \frac{[OH^-]}{[:B]_i} \times 100\%$$

	CH_3NH_2	+ H_2O	$\rightleftharpoons$ $CH_3NH_3^+$	+ OH^-
i	0.20		0	0
Δ	- 0.0094		+ 0.0094	+0.0094
eq	0.196		0.0094	0.0094

$$\%\text{ionization} = \frac{0.0094}{0.20} \times 100\%$$

$$= 4.7\% < 5\% \quad \text{OK}$$

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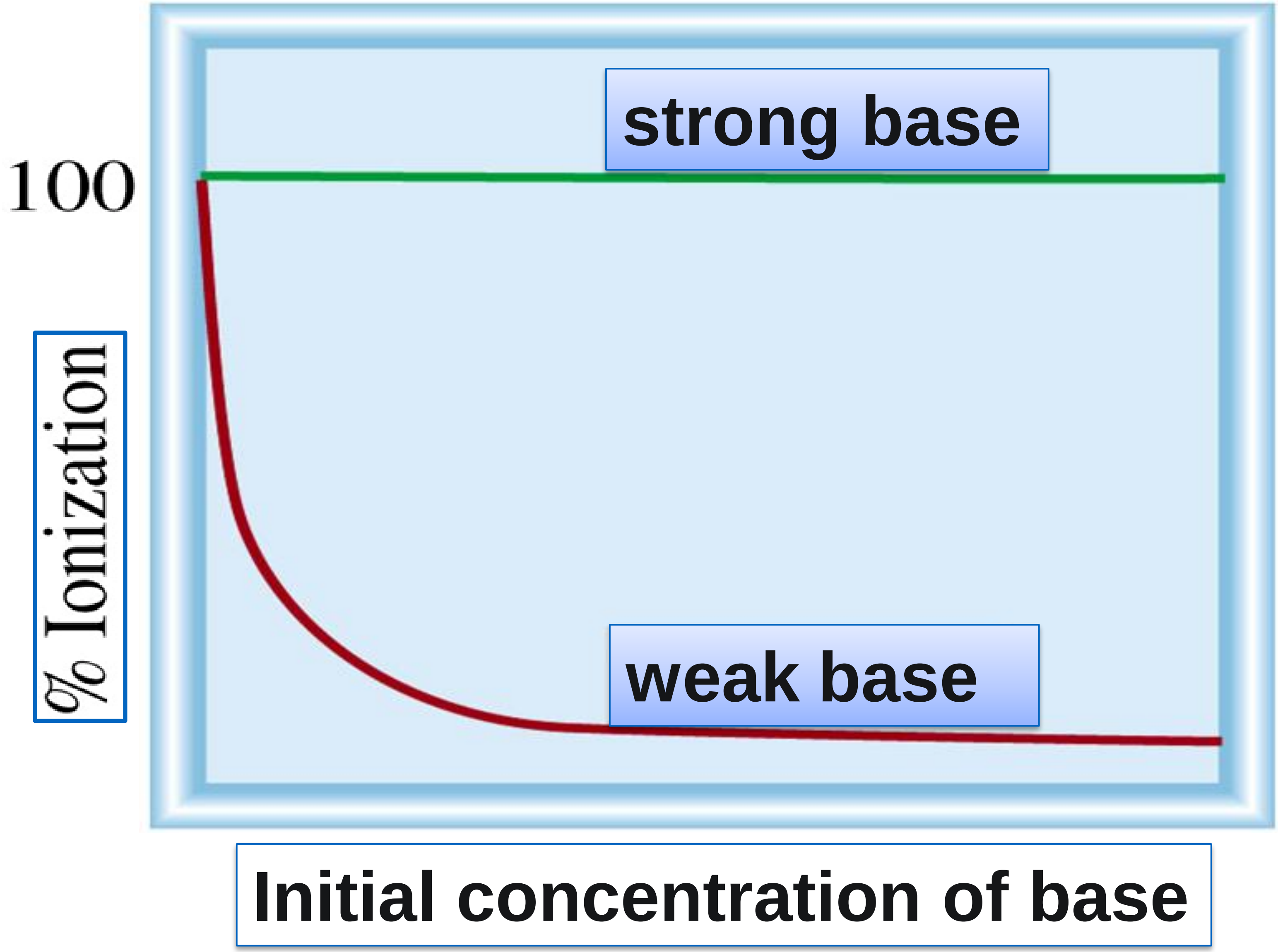
Percent Ionization

CH_3NH_2	0.40 *M*	0.20 *M*	0.020 *M*
$[OH^-]$	0.013	0.0094	0.00297
pOH	1.88	2.03	2.53
% ionization	3.3%	4.7%	14.8%

As the concentration of a weak base solution decreases, $[OH^-]$ decreases, the pOH increases, and the percent ionization increases



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A solution of NH_3 in water has a pH of 10.50. What is the molarity of the solution? $K_b = 1.8 \times 10^{-5}$.

	NH_3	+	H_2O	$\rightleftharpoons$	NH_4^+	+	OH^-
i	M				0		0
Δ	- x				+ x		+ x
eq	M - x				x		x

$$1.8 \times 10^{-5} = \frac{x^2}{M - x}$$

$$pOH = 14 - 10.50 = 3.50$$

$$x = [OH^-] = 10^{-pOH}$$

$$x = 10^{-3.50} \boxed{= 3.2 \times 10^{-4}\ M}$$

$$1.8 \times 10^{-5} = \frac{(3.2 \times 10^{-4})^2}{M - 3.2 \times 10^{-4}} \Rightarrow M = \boxed{6.0 \times 10^{-3}\ M}$$

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16.8 K_a-K_b Relationship

$$\cancel{NH_3} + H_2O \rightleftharpoons \cancel{NH_4^+} + OH^- \qquad K_b$$

$$\cancel{NH_4^+} + H_2O \rightleftharpoons \cancel{NH_3} + H_3O^+ \qquad K_a$$

$$H_2O + H_2O \rightleftharpoons H_3O^+ + OH^- \qquad K_w = K_a \cdot K_b$$

$$\boxed{K_a \cdot K_b = K_w = 1.0 \times 10^{-14}} \text{ at } 25\ ^\circ\text{C.}$$

➢ **The stronger the acid the weaker its conjugate base.**

$$K_a = \frac{K_w}{K_b} \qquad\qquad K_b = \frac{K_w}{K_a}$$

16.9 Acid-Base Properties of Salts

HCl ⇨ acidic

$NaCl$ ⇨ Neutral

$NaOH$ ⇨ Basic

CH_3CO_2Na ⇨ Basic

CH_3CO_2H ⇨ acidic

NH_3 ⇨ Basic

NH_4Cl ⇨ Acidic

- $NaCl$, CH_3CO_2Na and NH_4Cl are salts (MX)

$$MX \rightarrow M^+ + X^-$$ 100% ionization

NH_4Cl Solution

Salt ionization:

$$NH_4Cl(aq) \rightarrow NH_4^+(aq) + Cl^-(aq)$$

❖ **Hydrolysis:** Reaction of ions with H_2O to generate $OH^-(aq)$ or $H^+(aq)$

$$Cl^- + H_2O \rightarrow \text{ no reaction}$$

$$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+ \qquad \text{ACIDIC}$$

➢ If M^+ is conjugate acid for a weak base, its hydrolysis in water will produce H_3O^+ in solution, example NH_4^+.

Salt of weak base and strong acid: ACIDIC

CH_3CO_2Na Solution

Salt ionization:

$$CH_3CO_2Na(aq) \rightarrow Na^+(aq) + CH_3CO_2^-(aq)$$

Hydrolysis:

$$Na^+ + H_2O \rightarrow \text{no reaction}$$

$$CH_3CO_2^- + H_2O \rightleftharpoons CH_3CO_2H + OH^- \quad \textbf{BASIC}$$

- **If X^- is conjugate base for a weak acid, its hydrolysis will produce OH^- in solution, example NH_3.**

Salt of weak acid & strong base:	**BASIC**

NaCl Solution

Salt ionization:

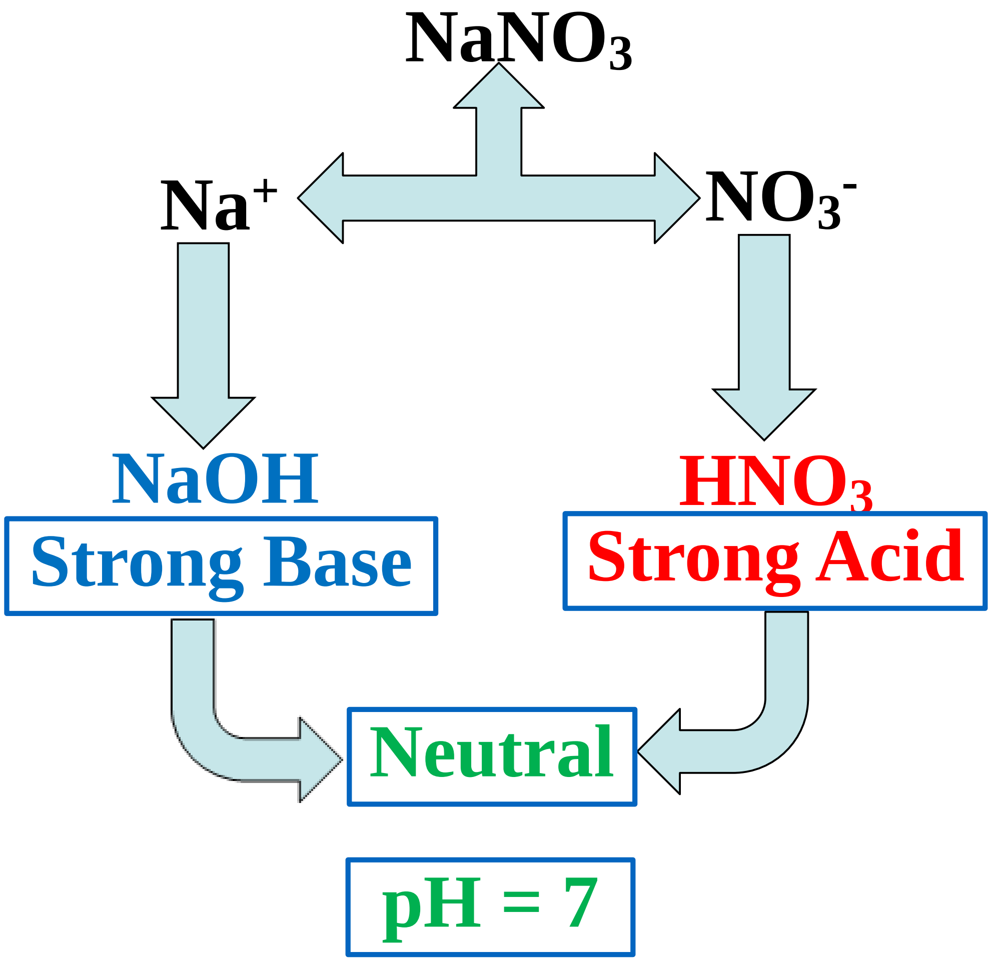
$$NaCl\ (aq) \rightarrow Na^{+}(aq) + Cl^{-}(aq)$$

Hydrolysis:

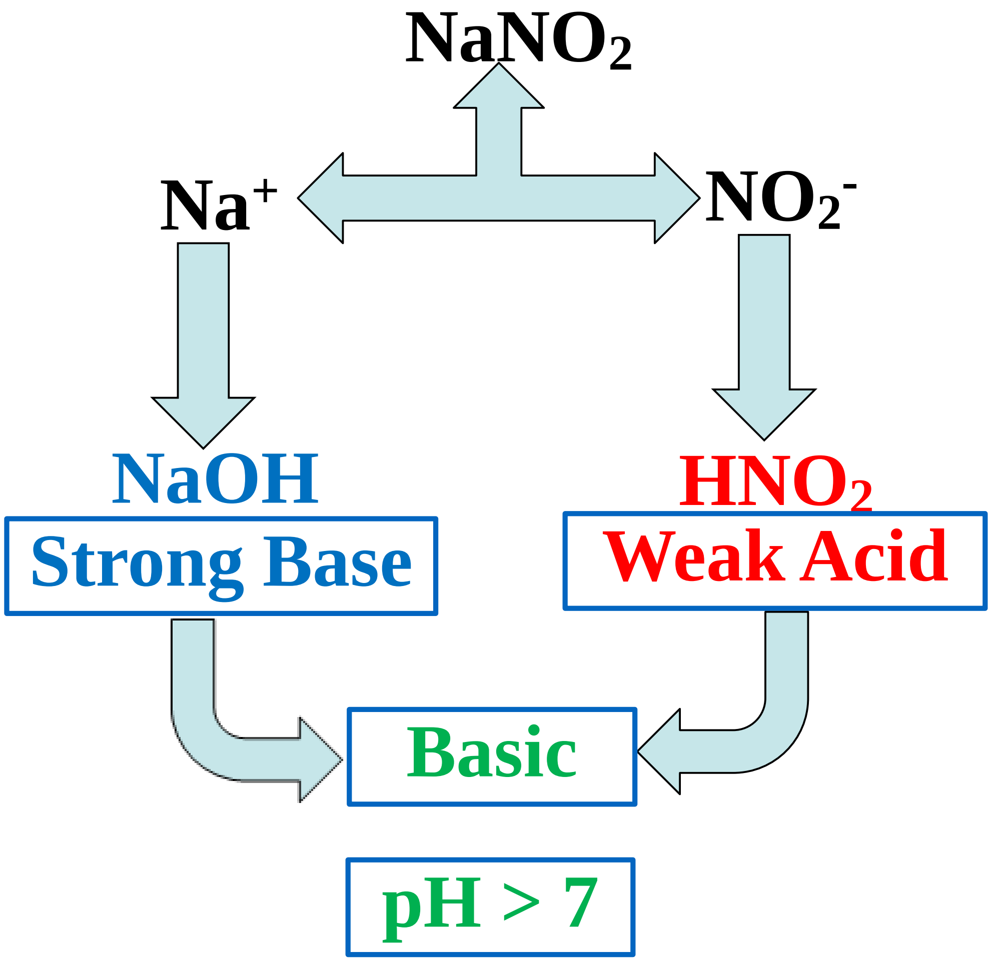
$$Na^{+} + H_2O \not\rightarrow NaOH + H^{+}$$ no reaction

$$Cl^{-} + H_2O \not\rightarrow HCl + OH^{-}$$ no reaction

Salt of strong acid & strong base: NEUTRAL

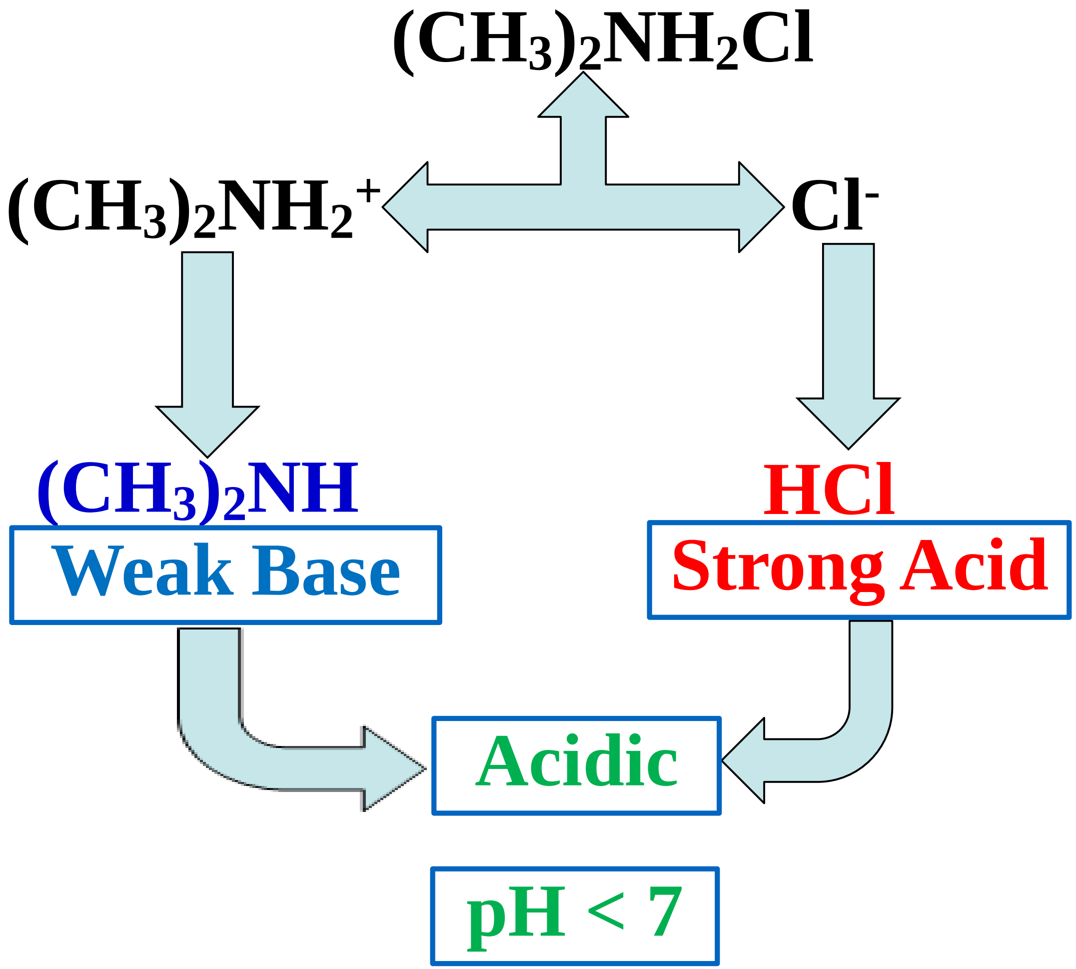


- ***Salts of Strong Acid-Strong Base Reactions***:
  $Ca(ClO_3)_2$, $BaClO_4$, LiBr, $KHSO_4$ etc.



- ***Salts of Weak Acid-Strong Base Reactions***:

  NaF, $NaNO_2$, $NaC_2H_3O_2$, etc.



➢ ***Salts of Strong Acid-Weak Base Reactions***:

NH_4Cl, NH_4NO_3, $(CH_3)_2NH_2Cl$, C_5H_5NHCl

NH_4NO_2

$K_a=5.6\times10^{-10}$ ⟸ NH_4^+ ⟸ ⟹ NO_2^- ⟹ $K_b=2.2\times10^{-11}$

NH_4^+ ⟹ NH_3 ; NO_2^- ⟹ HNO_2

$K_b=1.8\times10^{-5}$ ⟸ NH_3 — **Weak Base**

HNO_2 ⟹ $K_a=4.5\times10^{-4}$ — **Weak Acid**

depends on the relative strength of the acid and the base

$K_a>K_b$

Acidic

- ***Salts of Weak Acid-Weak Base Reactions*:**

  $NH_4C_2H_3O_2$, NH_4CN, NH_4NO_2, etc..

If $K_a>K_b$: **Acidic**	If $K_a<K_b$: **Basic**	If $K_a=K_b$: **neutral**

Ex: Calculate the pH of a 0.10 M KCN solution.

K_a (HCN) = 4.9×10^{-10}.

KCN

no reaction ← $H_2O + K^+$ $CN^- + H_2O \rightleftharpoons HCN + OH^-$

KOH

Strong Base

HCN

Weak Acid

Basic

pH > 7

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	$NaCN(aq)$	$\rightarrow$	Na^+	$+ CN^-$
i	0.10		0	0
Fi.	0.00		0.1	0.1

	CN^-	$+ H_2O$	$\rightleftharpoons HCN$	$+ OH^-$	K_b
i	0.10		0	0	
Δ	- x		+ x	+ x	
eq	0.10 - x		x	x	

$$K_b = \frac{K_w}{K_a} = \frac{1.0 \times 10^{-14}}{4.9 \times 10^{-10}} = 2.0 \times 10^{-5}$$

$$K_b = 2.0 \times 10^{-5} = \frac{x^2}{0.10 - \cancel{x}} \Rightarrow x = [OH^-] = 1.4 \times 10^{-3}\ M$$

$$pOH = -\log(1.4 \times 10^{-3}) = 2.85 \Rightarrow pH = 11.15$$

16.10 Acid-Base Behavior and Chemical Structure

❑ Factors that Affect Acid Strength

1. Bond Polarity.

H - X Na - H

δ+ δ- + -

Acid Proton donor Proton acceptor Base

2. Bond Strength

➢ The H-X bond must be weak enough to be broken.

3. Stability of conjugate base.

➢ The conjugate base, X^-, must be stable.

Sample Exercise

Rank the following binary acids according to their acidity.

HCl, HF, HBr, HI

$$\mathbf{H{-}F} < \mathbf{H{-}Cl} < \mathbf{H{-}Br} < \mathbf{H{-}I}$$

highest bond polarity

lowest bond polarity

strongest bond

weakest bond

bond enthalpy **567 > 431 > 366 > 299**

- The bigger the atom X the weaker the strength of the bond the more acidic the acid.
- Acid strength increases down a group.

Sample Exercise

Rank the following according to their acidity, H_2O, CH_4, HF, NH_3.

HF > H_2O >> NH_3 > CH_4

⇩ **highest bond polarity**

Highest acidity

⇩ **lowest bond polarity**

Neither an acid nor a base

- In a period the acidity strength increases with the increase in the **polarization** of the bond. That depends on the **electronegativity** of the element.
- The electronegativity difference between C and H is so small that the C-H bond is non-polar and CH_4 is neither an acid nor a base.
- In a period the acidity increases across a period {**from left to right**}.

Sample Exercise

Rank the following according to their acidity, H_2O, H_2S, H_2Se.

H_2Se > H_2S > H_2O

weakest bond | **Strongest bond**

highest acidity | **lowest acidity**

	GROUP			
	4A	5A	6A	7A
Period 2	CH_4 No acid or base properties	NH_3 Weak base	H_2O ---	HF Weak acid
Period 3	SiH_4 No acid or base properties	PH_3 Weak base	H_2S Weak acid	HCl Strong acid

Increasing acid strength (down)

Increasing base strength (up)

Increasing acid strength (to the right)

Increasing base strength (to the left)

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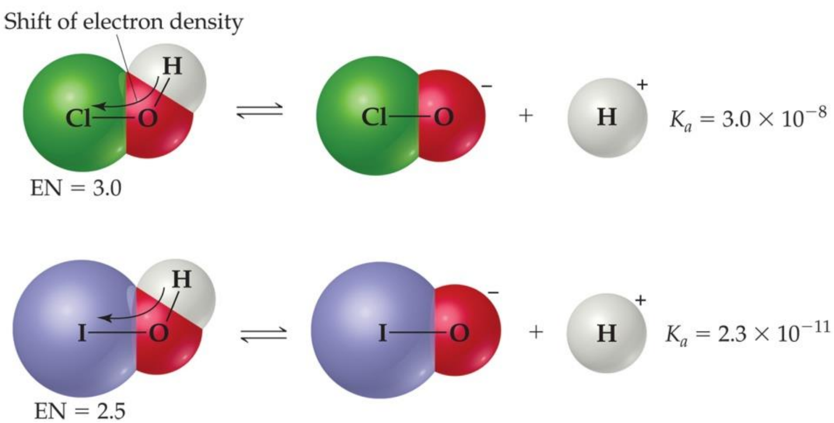
Oxyacids

Sample Exercise

Rank the following **Oxyacids** according to their acidity, HBrO, HClO, H_2O, HIO.

$$\overset{3.0}{\text{Cl-O-H}} > \overset{2.8}{\text{Br-O-H}} > \overset{2.5}{\text{I-O-H}} > \overset{2.1}{\text{H-O-H}}$$

Strongest acid — **weakest acid**



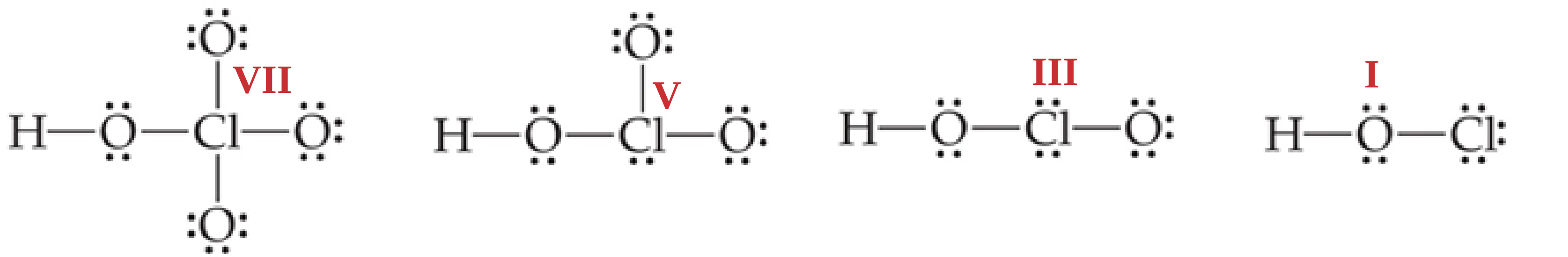
Acid	EN of Y	K_a
HClO	3.0	3.0×10^{-8}
HBrO	2.8	2.5×10^{-9}
HIO	2.5	2.3×10^{-11}

- In oxyacids, in which an OH is bonded to another atom, Y, the more electronegative Y is, the more acidic the acid.

Sample Exercise

Rank the following **Oxyacids** according to their acidity, $HClO$, $HClO_4$, $HClO_3$, $HClO_2$.

$$\mathbf{HClO_4 > HClO_3 > HClO_2 > HClO}$$



Perchloric	Chloric	Chlorous	Hypohlorous
strong acid	strong acid	$K_a = 1.1 \times 10^{-2}$	$K_a = 3.0 \times 10^{-8}$

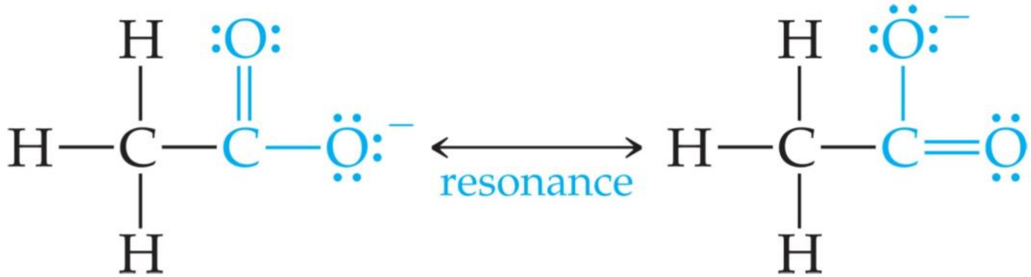
- Increasing the number of oxygen atoms, stabilize the conjugate base by resonance.

➢ In a series of oxyacids, the acidity increases as the number of oxygen atoms increase {as the oxidation number of the central atom increases}.

$$\mathbf{HBrO_4 > HBrO_3 > HBrO_2 > HBrO}$$

Sample Exercise

Rank the following **Carboxylic Acid** according to their acidity, CH_3CO_2H, CH_2ClCO_2H, $CHCl_2CO_2H$, CCl_3CO_2H.



- Resonance in the conjugate bases of carboxylic acids stabilizes the base and makes the conjugate acid more acidic.

$$CCl_3CO_2H > CHCl_2CO_2H > CH_2ClCO_2H > CH_3CO_2H$$

- The strength of carboxylic acid increase as the number of electronegative atoms in the acid increases.

16.11 Lewis Acids & Bases

$H^+ \quad + \quad :NH_3 \rightarrow NH_4^+ \Rightarrow$

$$\begin{array}{ccc} & H & \\ & | & + \\ H - & N & - H \\ & | & \\ & H & \end{array}$$

H^+: electron pair ($2e^-$) acceptor → **Lewis Acid**

$:NH_3$: electron pair ($2e^-$) donor → **Lewis Base**

$$:NH_3 + BF_3 \rightarrow H_3N - BF_3$$

$:NH_3$: **Lewis Base** | BF_3: **Lewis Acid**

$$6H_2O + Al^{3+} \rightarrow Al(H_2O)_6^{3+}$$

$6H_2O$: **Lewis Base** | Al^{3+}: **Lewis Acid**

Ex: **Identify the Lewis acids and Lewis bases in the following reactions:**

(a) $H_2O + CO_2(aq) \rightarrow H_2CO_3(aq)$

(b) $Cu^{2+}(aq) + 4NH_3(aq) \rightarrow Cu(NH_3)_4^{2+}(aq)$

(c) $AlCl_3 + Cl^- \rightarrow AlCl_4^-$

Hydrolysis of Metal Ions

$$Fe^{3+} + 6\ H_2O \rightleftharpoons Fe(H_2O)_6^{3+}$$ hydration

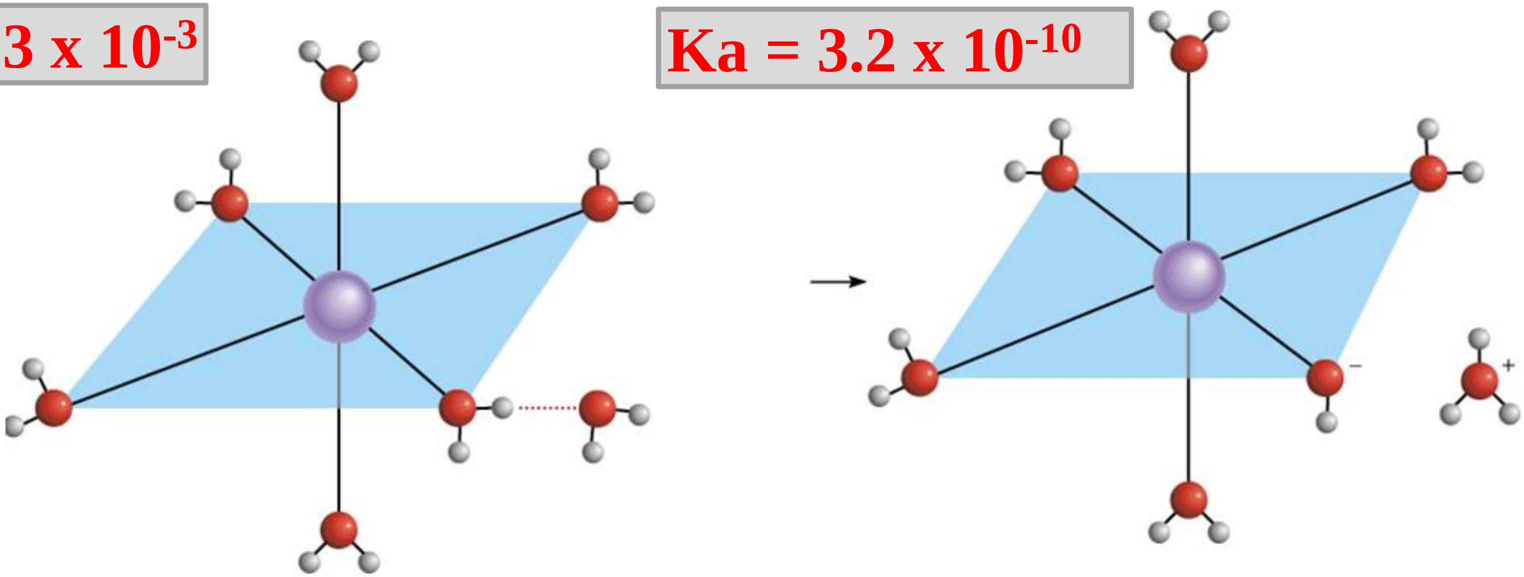
L. Acid | L. Base

$$Fe(H_2O)_6^{3+} + H_2O \rightleftharpoons Fe(H_2O)_5(OH)^{2+} + H_3O^+$$

Acid | Base

$K_a = 6.3 \times 10^{-3}$

$K_a = 3.2 \times 10^{-10}$



- K_a generally increases with Increasing metal charge

1 *M* soln. of	$Al(NO_3)_3$	$Zn(NO_3)_2$	$Ca(NO_3)_2$	$NaNO_3$
pH	3.5	5.5	6.9	7

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